Numerical Solution of the 2D Stokes Equations

Staggered Grid Discretization

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1 Introduction

We consider the two-dimensional Stokes equations

$$\begin{aligned} \mu\Delta \boldsymbol{u} - \nabla P &= \boldsymbol{b}, \\ \nabla \cdot \boldsymbol{u} &= 0, \end{aligned} \tag{1}$$

Where $\boldsymbol{u}$ is the velocity vector of the fluid, defined as $\boldsymbol{u} = [U(x,y), V(x,y)]$, whose components are the x and y components of the velocity, respectively. We also assume $\mu = 1$ and we impose periodic boundary conditions for U, V, P on the left and right boundaries, Dirichlet conditions for U in the y direction where $U = 0$ on top and bottom boundaries, and Dirichlet conditions for V in the y direction, where $V = -3.5$ units on our top and bottom boundaries. We discretize the equations on a staggered (MAC) grid and write the discrete unknowns and their boundary conditions as a sparse saddle point system.

2 Numerical Method

We use the method of manufactured solutions to solve the problem. We are given:

$$\begin{aligned} U_{an}(x,y) &= \sin(2\pi y)\sin(2\pi x), \\ V_{an}(x,y) &= -3.5 + \cos(2\pi x)(\cos(2\pi y) - 1), \\ P_{an}(x,y) &= \sin(2\pi y)\cos(2\pi x). \end{aligned}$$

We can now compute the forcing functions. We get:

$$\begin{aligned} f(x,y) &= (-8\pi^2 + 2\pi)\sin(2\pi y)\sin(2\pi x), \\ g(x,y) &= \cos(2\pi x)[4\pi^2 - (8\pi^2 + 2\pi)\cos(2\pi y)]. \end{aligned}$$

We can thus let $N_x = N_y = N$ and $h = 1/N$ and construct the MAC grid where $U_{i,j}$ is stored at coordinates $(ih, (h-1/2)h)$, $V_{i,j}$ stored at $((i-1/2)h, jh)$, and $P_{i,j}$ stored at $((i-1/2)h, (j-1/2)h)$. We have pressure located at cell centers, x velocity on left and right faces of pressure cells, and y velocities on top and bottom faces of pressure cells.

We then construct

$$\begin{bmatrix} \mu\mathcal{L}_u & 0 & -G_x \\ 0 & \mu\mathcal{L}_y & -G_y \\ D_x & D_y & 0 \end{bmatrix}$$

in matlab, with prescribed velocity values moved to the right hand side of the system at boundaries. Since pressure is not unique, we prescribe the pressure at one grid point $P_{1,1} = 0$. We then use backslash to solve.

3 Results and Error Analysis

The following graph is the solution plot of the magnitude of $\boldsymbol{u}$ attained from solving the system in Matlab using backslash:

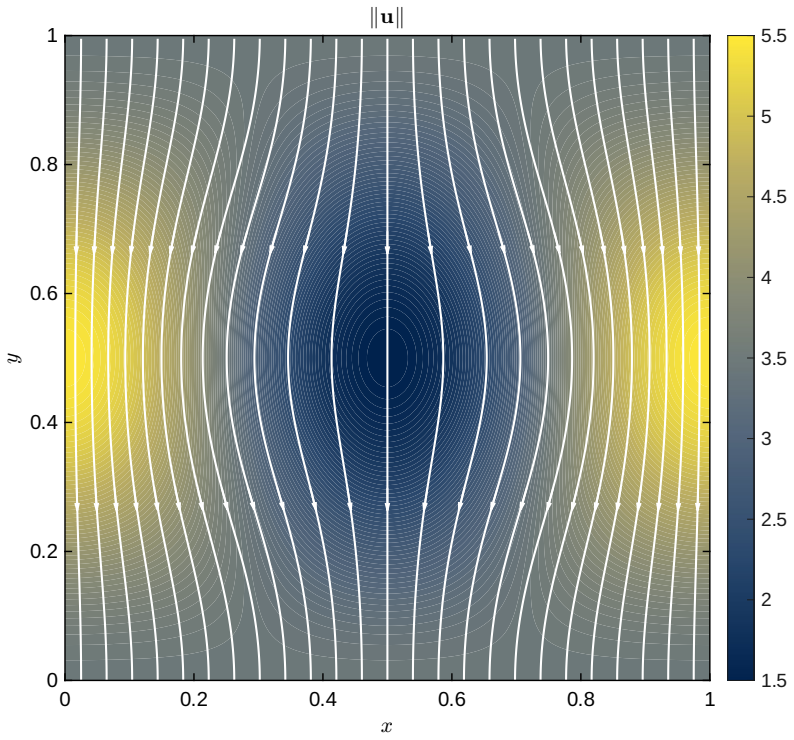


Figure 1: Solution plot of magnitude of $\boldsymbol{u}$

Figure 1 illustrates the magnitude of $\|\boldsymbol{u}\| = \sqrt{U^2 + V^2}$. The streamlines indicate the direction of fluid flow. The flow is downward, consistent with prescribed boundary condition for the vertical velocity, $V = -3.5$. The velocity magnitude is smallest near the center, while reaching a maximum near the x boundaries.

The relative error is defined by

$$E_U = \frac{\|U_{\text{num}} - U_{\text{an}}\|_2}{\|U_{\text{an}}\|_2}. \tag{2}$$

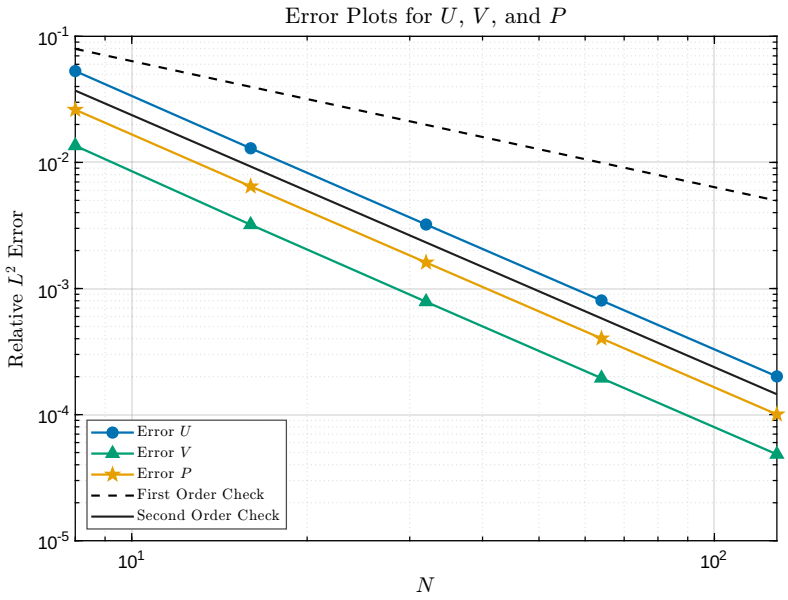


Figure 2: Relative error versus grid resolution.

Figure 2 graphs the L^2 errors for U, V, and P as the N grows. All errors decrease with increasing N, roughly proportional to $1/N^2$, as can be seen by comparing the slopes of the second order check with the slopes of the errors of U, V, and P. This confirms that our MAC discretization is a second order approximation for the manufactured solution.

4 Additional Experiment

5 Conclusion